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RESEARCH ARTICLE



Explaining change in independence–unification views in Taiwan: a within-between model

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ABSTRACT

Independence–unification (IU) views have been profoundly influential in Taiwan politics. This study uses the rally ‘round the flag effect to explain the increase in support for Taiwan independence after President Tsai Ing-wen rejected China’s ‘one country, two systems’ proposal. Taking advantage of within-individual differences in panel data, we can estimate the probability of IU views given a set of certain characteristics compared with other values of those characteristics and ascertain the influence of change in variables such as approval of the president’s handling cross-Straits relations at different time points. Our preliminary results show that Taiwanese people indeed change their IU views due to the external and domestic environment. People who perceive that the economy has improved and whose views on President Tsai have become more favorable tend to flock towards the independence side. People whose views of President Tsai’s handling of cross-Straits relations have become more favorable also move towards independence. If we only consider differences between individuals, feelings about the KMT and DPP stand out as significant predictors of IU views.

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Introduction

Independence–unification (IU) views have been profoundly influential in Taiwan politics. Such views reflect Taiwanese people’s perceptions of current and future relations between Taiwan and China. Like most political issues, they are often subject to predispositions but may also respond to important events. Conventional regression models that mostly use cross-sectional data only consider between-person variation and cannot detect the change *within* individuals. In this case, we are not able to capture the dynamics of IU views, which are critical not only to geopolitics in Asia but also to research on public opinion in Taiwan.

After the Chinese Civil War fought between the Kuomintang (KMT), also referred to as the Chinese Nationalist Party, and the Chinese Communist Party (CCP), the CCP established the People’s Republic of China (PRC) in mainland China and the defeated

KMT-led government of the Republic of China (ROC) fled to Taiwan. During the 1990s, the KMT-led authoritarian regime in Taiwan underwent a transition to democracy. Free and fair elections, political pluralism, and low corruption now embody Taiwan's democracy. The opposition movement mobilized supporters through Taiwanese identity (Lin, 1989, 2002). They argued that the Taiwanese people have the right to self-determination and claimed that if Taiwan declared independence and became a nation, China would no longer have the right to claim that Taiwan is part of China and the KMT would lose its legitimacy. In 1986, the Democratic Progressive Party (DPP) was established with the pursuit of sovereignty and statehood at the center of its platform. Conversely, the KMT maintained the position that the Republic of China is the only legitimate government of China. The KMT also argued that due to Taiwan's relatively small size, it would benefit from reunification with the mainland due to the shared cultural heritage of both sides. In summary, the relations between China and Taiwan are very complicated, and Taiwanese people's views on the IU spectrum play an important role in this issue.

Previous studies either concentrate on the variation in IU views among the Taiwanese population or its evolution over time. So far, there are no studies that address the within- and between-individual effects on IU views. To fill this gap, we applied the 'rally 'round the flag' effect, that is, presidential approval is boosted during international crises (Mueller, 1973; Oneal & Bryan, 1995; Tsai, 2000). According to this argument, when the nation is threatened, incumbents benefit from citizens' empathy towards their leaders. In the case of Taiwan, when cross-Strait relations become tense, Taiwanese people may unite behind the president and show support for the regime's cross-Strait policy. Therefore, we argue that respondents' evaluations of the president's handling of cross-Strait relations can represent their reactions to cross-Strait relations in both the long run and the short run. If they approve of the way the president handles cross-Strait relations, they will, in turn, support the president's views on this issue, which lean towards independence. By the same token, we also examine the effects of economic fluctuation, presidential approval, and partisanship on voters' IU views in both the long run and the short run. There are two important implications of the empirical analysis. First, the same event might have both short-term and long-term effects on voters' IU views. Second, the short-term and the long-term effects might be different and even contradictory.

In order to estimate attitude change, it is necessary to distinguish short-term effects from long-term effects. Therefore, the present study adopted a within-between model to analyze panel data. This approach has been applied to research in communications, sociology, and psychology. For example, cross-sectional data analysis has provided evidence that media consumption may influence people's acceptance of vaccines (Piltch-Loeb et al., 2021). By using panel data and the within-between model, it is possible to estimate the effect of change in media consumption on intention to take the vaccine.

The remainder of this article proceeds as follows. The section 'Evolution of Independence–Unification Views' introduces the general background of IU views. The section 'Previous Literature' reviews the extant literature and develops our arguments with testable hypotheses. The section 'Data, Variables, and Statistical Models' illustrates the data, measures, and model specifications. The section 'Main Results' presents the descriptive statistics of the main variables and the results of survey data analysis. The section 'Conclusion' concludes the article.

Evolution of independence–unification views

Social cleavages are the foundation of party systems. According to Lipset and Rokkan (1967), party development reflects social cleavages derived from nation-building and the Industrial Revolution, such as religion, social class, and the urban-rural divide. Voters belonging to different social groups aligned with political parties, and party organizations developed around social cleavages. These developments stabilized party systems in advanced industrial societies until the 1960s. During this process, political identities emerged, and ‘internally created’ parties became prominent in parliaments (Mair, 1997, p. 185). Since the 1970s, the value-based ideologies have structured party systems around non-economic issues (Bornschier, 2010). In particular, the rise of post-materialism has weakened the social class-based dimension of group polarization (Inglehart, 1984).

As for Taiwan, the longstanding independence–unification issue remains the most salient division between parties (Achen & Wang, 2017; Sobel, Haynes, & Zheng, 2010). As in other countries, the past authoritarian regime has contributed to the stability of party systems in Taiwan (Hicken & Kuhonta, 2015). In the 1990s, the KMT and DPP competed on the scope and pace of democratization. During the passage of a series of constitutional amendments to consolidate the presidential system, the national identity debate also gradually emerged as the new axis of party competition. In addition, scholars have shown empirical evidence that the independence–unification issue plays a dominant role in party competition in Taiwan. In a study of the 1992 legislative election, Hsieh and Niou (1996) estimated the impact of four issues on voting behavior, including national identity (referring to the independence–unification issue), and found that national identity had a significant impact on party evaluation and voter choice. Hsieh (2005) also demonstrated that national identity can predict party identification. Sheng and Liao (2017) examined four issues measured on a 0–10 point scale between 1996 and 2012. The authors found that IU is the most salient issue because of the wide gap between KMT and DPP partisans on this issue. Cheng and Hsu (2015) argued that the IU issue has defined questions of economic and trade relations between China and Taiwan. Achen and Wang (2017, p. 13) also argued that ‘the current central issue of politics in Taiwan is the issue of unification vs. independence.’

While most existing research confirms the salience of IU views, few studies answer this question: are people’s IU views constant or movable over time? Tsai, Wang, and Tossutti (2008) investigated the stability of individual policy preferences between 2001 and 2004. Drawing on panel data from the Taiwan Election and Democratization Study (TEDS) and using ordered probit modeling, they found national identity and evaluations of national economy influence IU views. They also showed that the IU issue trended from a pro-unification position towards the status quo. Using pooled six cross-sectional data and age-period-cohort analysis, Huang (2019a) argued that generational replacement may account for the swing in attitudes toward pro-independence views. However, his conclusions are based only on data covering the period between 1996 and 2016. More importantly, Huang used cross-sectional instead of panel data, which may avoid panel attrition problems but relies on a small set of common variables shared by all of the cross-sectional surveys.

Following Tsai et al. (2008) and Huang (2019), we assume that there are long-term and short-term factors influencing IU views. Demographic background and longstanding attitudes may be responsible for the variation in IU views between individuals, while events may cause individuals to change their IU views in the short term. Wu (2002) identified three types of IU position: Taiwan nationalism, Chinese nationalism, and pragmatism. He argued that Taiwan nationalism is fluid and that ethnic identity (Taiwanese, Chinese, or both) is a better predictor of nationalism. Keng, Chen, and Huang (2006) also noted the importance of material interest and political identity and found that attitudes toward cross-Straits economic integration are largely decided by political identity rather than self-interest. Hsu and Chen (1998) proposed a different approach to this question, arguing that highly-educated people tend to change their position if they receive information contrary to their attitudes. Their research provides evidence for the rational component of IU views.

Another cluster of literature focuses on attitude stability and change. Examining 1958–1960 panel data, Converse (1964) found that respondents show relatively lower consistency on issues regarding housing and economic aid. He pointed out that the mass public lacks information and logical thinking so they merely make sense of concrete, simple subjects. According to Converse and Markus (1979) and Markus (1986), although people's attitudes are changeable, they often wrongly perceive their positions in the past. Krosnick (1991) studied panel data from the 1960s and 1970s, concluding that preferences over social groups and non-racial domestic issues appeared to be as stable as partisanship and ideology. Mutz and Dilliplane (2011) explored the effects of change in perceived distance on a feeling thermometer of several presidential candidates between 2007 and 2008. Using a within-subject design, the authors estimated a fixed-effect regression model and showed that decreasing the ideological distance from the potential voter may benefit electoral prospects.

With the exceptions of Tsai et al. (2008) and Huang (2019a), the wisdom ascertained from most of the previous literature concerning IU views comes from cross-sectional data. We instead try to model short-term changes in IU views using rare panel data. Since panel data is scarce, we are fortunate to be able to test our hypotheses using this newly-collected panel data. Unlike Mutz and Dilliplane (2011), we use the within-between model because it can distinguish between short- and long-term effects.

We use the rally 'round-the-flag effect to explain attitude change. Mueller (1973, p. 21) reported that a president's approval rating increases during events that (1) are international, (2) involve the United States and the president directly, and (3) are specific, dramatic, and sharply focused. Mueller (1973, p. 212) concluded, 'the public seems to react to "good" and "bad" international events in about the same way.' The rally 'round the flag theory has been widely tested and accepted (Hetherington & Nelson, 2003; Oneal & Bryan, 1995). In terms of the causes of the rally 'round the flag effect, Lee (1977, p. 253) pointed out that 'the public tends to support a presidential action (or inaction at times) regardless of its content, when there is not enough information of evaluating the situation.' Hetherington and Nelson (2003) suggested that international events boost patriotism, which can be characterized as the negative views of an in-group toward out-groups. Furthermore, Lambert, Schott, and Scherer (2011) emphasized that events can influence policy preferences and presidential approval. They argued that 'As such, attacks against the in-group can certainly instigate anger, which might

facilitate aggressive action tendencies.’ Outside the United States, Wang and Cheng (2021) have identified the rally ‘round the flag effect in Taiwan, arguing that the public’s support for the president is contingent on whether people are satisfied with the president’s handling the COVID-19 crisis.

Furthermore, we assume that media exposure is related to IU views. There has been a great deal of research on the relationship between media exposure and political attitudes (Campante & Hojman, 2013; Della Vigna & Kaplan, 2007; Gentzkow & Shapiro, 2004; Prior, 2007). People often select media according to their party preferences, and media content may reinforce their predispositions. Therefore, political polarization is related to media exposure. In Taiwan, partisan bias is a glaring problem in television news (Lo, King, Chen, & Huang, 1996). It has also been shown that political news use impacts political participation (Zhang, 2012). Therefore, we propose that IU views are associated with sources of political news. Newspapers and television are both very important news sources, with television news gaining greater popularity due to its accessibility to audiences (Li, 2001, p. 268). The penetration rate of cable TV was 81.6% in 2006, and 24-hour TV news programs have a large influence in Taiwanese politics (Hsu, 2014). According to the Media Book published by the Media Agency Association, in 2020, 84.2% of respondents watched television while only 18.9% of respondents read newspapers.¹ Therefore, we focused on television news and measured people’s media exposure by asking them to list all of the pro-KMT or pro-DPP television news channels that they watch for political news.

Drawing on previous literature, we assume that dramatic international events may trigger people’s fear of out-groups and emotional identification with the regime, i.e. rally ‘round the flag. In addition to presidential approval, the rally effect may also happen in other areas, such as general trust in government and support for political institutions (Hetherington & Nelson, 2003; Parker, 1995). In this regard, Taiwanese people perceived Xi’s ‘one country two systems’ proposal as a threat against Taiwan. Therefore, Tsai’s rebuttal of Xi’s proposal boosted support for Taiwan independence, one of the longstanding policy areas owned by the DPP.

Research hypotheses

Figure 1 shows a dramatic increase in pro-independence support in 2019 and 2020.² The combined proportion of ‘immediate independence’ and ‘independence later’ (two solid lines) increased from 20.7% to 25% in the first quarter of 2019. At the same time, the combined proportion of ‘immediate unification’ and ‘unification later’ (two dot-dashed lines) dropped from 14.6% to 11.9%, while ‘status quo forever’ and ‘maintain the status quo, decide later’ remained almost the same. In the second quarter, ‘unification’ contracted to 7.9% while ‘independence’ grew to 29.7%. Although the two independence categories fell slightly in the first quarter of 2020, they rebounded in the second quarter. As of June 2020, about 35% of respondents preferred ‘immediate independence’ or ‘independence later.’ In other words, one-third of Taiwan’s people agree that Taiwan should become a sovereign state even though China threatens the use of force if Taiwan declares independence.

Why has pro-independence support increased since the last quarter of 2018? The first explanation is President Tsai’s rejection of Xi Jinping’s ‘one-country, two-systems’

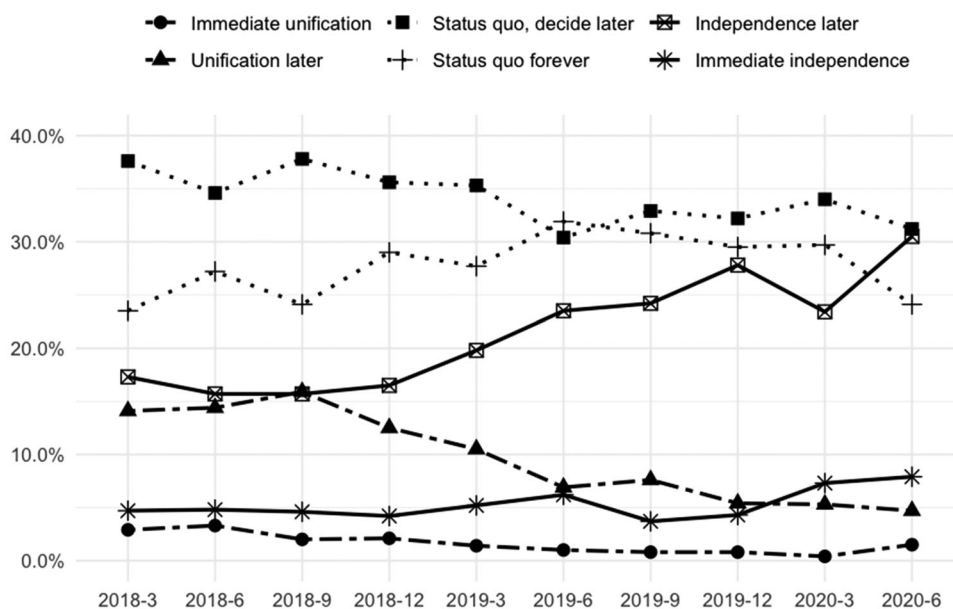


Figure 1. Trends on the Independence–Unification Issue Position, 2008–2020. Data: Huang (2020).

proposal in January 2019. About one month before Xi's proposal, the DPP was defeated by the KMT in the 2018 local elections, dropping seven mayorships and even losing one of their strongholds, Kaohsiung City.³ Tsai seized on Xi's proposal as the opportunity to frame the cross-Straits relations as the most prominent issue, highlighting China's hostility and threat to Taiwan's democracy. Alongside the salience of cross-Straits relations, protests against a proposed extradition bill broke out in Hong Kong in June, which further increased fear and loathing of the Chinese government in Taiwan. Although China did not repeat the rhetoric of 'one country, two systems,' the 'sense of the nation's impending doom,' or fear of becoming part of China, has become widespread among voters, contributing to the defeat of the KMT in the 2020 presidential election.⁴

The other explanation is the improvement of Taiwan's domestic environment in the past two years. Prosperity may encourage people to value autonomy. Due to the US–China trade war that began in 2018, many Taiwanese companies that previously exported network appliances, servers, and other electronic products to the U.S. from China increased their production lines in Taiwan to avoid high tariffs imposed by the U.S. According to statistics from the Directorate General of Budget, Accounting and Statistics (DGBAS), Taiwan's GDP growth rate in the first three quarters of 2019 was 1.84%, 2.60%, and 2.99% respectively, while in the fourth quarter it rose to 3.38%. Although the COVID-19 pandemic that erupted in January 2020 subdued tourism, food service, and retail sales, the annual growth rate was 3.11% in 2020, ahead of Hong Kong, Singapore, and South Korea.⁵

The DPP administration's response to COVID-19 may strengthen Taiwanese people's resolve against China, which has ordered the World Health Organization to exclude Taiwan (Wong, 2020). Following the first outbreak in Wuhan, China before the Lunar New Year, Taiwan not only implemented strict border quarantine measures, but also

traced people with low risk who had no history of travel to level 3 alert areas (Wang, Ng, & Brook, 2020). Furthermore, the Taiwan government mobilized local industry to boost the supply of face masks. After ensuring that the quantity of masks was sufficient to supply the local population, Taiwan donated millions of masks to countries hit hard by the coronavirus, including the United States and many European countries. American and European officials and celebrities praised the self-ruled but isolated democracy as an example of how to successfully fight against COVID-19 with less coercive measures (Hernández & Horton, 2020).

While individuals may respond to current issues and shift their IU views, we should note that the IU dimension reflects generational difference and party loyalty. Generational replacement has a longstanding effect on the distribution of IU views (Huang, 2017). Most young cohorts under 40-years-old prefer independence partly because they have the experience of democratic transition including the end of martial law and emergence of opposition parties when they became eligible to vote for the first time (Lin, 2015). Since the 2000s, the two main parties, KMT and DPP, have confronted each other on cross-Straits relations and underlying national identity (Schubert, 2004). Although Lee Teng-hui, the late KMT chairman and president, attempted to 'indigenize' the KMT under the 'one China, two entities' framework, the party ousted Lee after his successor, Lien Chan, lost his bid for the presidency in 2000. Since then, the KMT has spoken positively about re-unification with China. On the contrary, the DPP has promoted Taiwan nationalism as a sense of political community distinct from China. The KMT and DPP continue to disagree on cross-Straits policy, and voters also recognize the difference between the parties on the IU spectrum (Sheng & Liao, 2017). Therefore, age cohort and partisanship are associated with changes in IU views.

Based on our literature review, we propose the following hypotheses:

Hypothesis 1A: [Short-term Evaluations of Cross-Straits Relations] Satisfaction with President Tsai's handling of cross-Straits relations influences IU views.

Hypothesis 1B: [Long-term Evaluations of Cross-Straits Relations] Satisfaction with President Tsai's handling of cross-Straits relations influences IU views.

Between 2018 and 2019, tensions grew between China and Taiwan after Xi called on Taiwanese to accept 'one country, two systems' as a model for reunification, leading Tsai to strike back instantly. People who agree with Tsai's response may shift their IU views towards independence, but people who have different opinions about Tsai's policy may move towards unification or maintain their unification views. Therefore, Hypothesis 1A states that within-individual change in the evaluations of Tsai's cross-Straits policy may account for respondents' IU views in 2019. Conversely, Hypothesis 1B indicates that there is a long-term temporal relationship between approval of Tsai's handling of cross-Straits relations and IU views. President Tsai has long insisted that Taiwan is a sovereign state and refused to accept the 'one China principle,' leading the KMT to criticize her for provoking China and DPP supporters to commend her for consolidating Taiwan as a nation. Therefore, we assume that people who disagree with Tsai's handling cross-Straits relations are more likely to lean toward unification than other people.

Hypothesis 2: [Economic evaluations] Retrospective economic evaluations have an effect on IU stance.

Voters hold the government accountable for economic performance in democracies, and Taiwan is no exception (Tsai, 2017). Because China is Taiwan's largest export partner, cross-strait relations are involved with economic development. The KMT returned to power in 2008 partly because their 'peace bonus' campaign was very successful. Tan and Ho (2017) found that one's IU stance is directly correlated with economic evaluations – independence supporters tend to claim that economy is worse when the KMT is in power. As voters perceive that economy is not improving, they may switch to different views of cross-strait relations held by the opposing camp. Therefore, we assume that voters prefer independence if they evaluate the state of the economy positively.

Hypothesis 3: [Presidential Approval] Presidential approval influences independence–unification views regardless of time periods.

Hypothesis 3 notes that presidential approval influences people's IU views. If the president is popular, people are more likely to trust and follow the leadership of the president. In other words, people who approve of President Tsai's performance are likely to support either independence or the status quo because she refuses to accept the '92 Consensus,' which implies that Taiwan and PRC agree that there is 'one China' but the definition of 'China' is subject to each side's interpretation as a precondition of interaction by China.

Hypothesis 4: [Media exposure] The number of pro-KMT and pro-DPP television news channels viewed is related to IU views.

Since there is partisan bias in Taiwan's television news, we assume that exposure to partisan news channels is related to IU views. Following Dilliplane, Goldman, and Mutz (2013), we count the number of pro-KMT and pro-DPP news channels that each respondent often watches. In doing so, we construct two continuous variables and expect that the number of pro-DPP and pro-KMT news channels watched is related to changes in IU views.

Hypothesis 5: [Partisanship] DPP identifiers are more likely to hold independence views and KMT identifiers are more likely to choose unification.

Since the 1990s, the DPP and KMT have been placed on the pro-independence and pro-unification sides of Taiwan's political spectrum (Hsieh & Niou, 1996). While Tsai is relatively moderate on the issue of national status, sovereignty and independence remain the DPP's core values. On the contrary, the KMT emphasizes the '92 Consensus' as the cornerstone of cross-strait relations. Therefore, we assume that either long-term or short-term KMT partisans are likely to prefer unification and that DPP partisans choose independence.

Data, variables, and statistical models

We analyzed panel data in Taiwan with a nationally representative sample. The panel survey was conducted by the Taiwan Institute for Governance and Communication Research (TIGCR) sponsored by the Ministry of Education and led by Chi Huang and Ching-ching Chang. The first wave was collected through face-to-face interviews between 7 August and 23 November 2018 and consists of 2484 respondents. Of the

first wave respondents, 1388 respondents were re-interviewed in the second wave between May and October, 2019. Before conducting interviews, TIGCR sent the survey materials to the Institution Review Board (IRB) of National Chengchi University (NCCU) for human research review. The NCCU IRB granted permissions for both survey waves.

Our dependent variable was a 6-point scale of IU views, in which 1 represents ‘immediate unification’ and 6 ‘immediate independence.’ The ordering ‘immediate unification,’ ‘unification,’ ‘status quo, decide later,’ ‘status quo forever,’ ‘independence,’ and ‘immediate independence’ has been widely accepted by scholars, including Sobel et al. (2010) and Tan and Ho (2017). In Hsiao and Yu’s (2012) re-examination of the 6-itemed measurement, people who answer ‘status quo forever’ are actually closer to the pro-independence side and people who choose ‘status quo, decide later’ are closer to pro-unification camp. Therefore, we use this measurement as a continuous variable.

Our main independent variables are:

- (1) Evaluations of national economy: We measured retrospective evaluations of the national economy by asking respondents: ‘Compared to one year ago, do you think Taiwan’s economy is better, the same, or worse?’ Responses are classified as ‘much worse,’ ‘worse,’ ‘the same,’ ‘better,’ and ‘much better.’
- (2) Evaluations of President Tsai’s handling of cross-Strait relations. We asked respondents to rate the way Tsai deals with cross-Strait relations with four categories: ‘very dissatisfied,’ ‘dissatisfied,’ ‘satisfied,’ and ‘very satisfied.’
- (3) Evaluations of Tsai’s performance. Respondents were asked to evaluate Tsai’s overall performance with four categories: ‘very dissatisfied,’ ‘dissatisfied,’ ‘satisfied,’ and ‘very satisfied.’
- (4) Media exposure. Respondents were asked to list all of the television news channels that they often watch. Based on the report on the content of television news released by the National Communication Commission (2019) and Hsu’s analysis (2014), we counted the number of pro-KMT and pro-DPP TV channels listed by each respondent. For more details on the classification scheme, please see Appendix A.
- (5) Partisanship: Partisanship was measured by the following question: ‘Political parties in Taiwan have their own supporters. Among these parties, which one do you support?’

We also included demographic background (gender, age, and education) in the model to check whether any of them has partial correlation with the outcome variable. Furthermore, the between-effect of Taiwanese or Chinese identity was controlled for because it is the center of Taiwan politics (Achen & Wang, 2017). For more information about variable, please refer to Appendix A.

The within-between regression model

We are interested in both individual and temporal effects on IU views. To separately estimate these two effects, we applied within-between random effects (RE) modeling. RE models – also called multilevel models, hierarchical models and mixed models – have

been increasingly used in a variety of disciplines due to the concern over cross-sectional and temporal variation. For an illustration of the statistical models, suppose that there exists panel data with a dependent variable $y_{j,t}$ for unit $j = 1, \dots, J$ measured at time $t = 1, \dots, T$. a standard RE model would be (Gelman & Hill, 2007):

$$y_{j,t} = \beta_{0,j} + \beta_1 x_{j,t} + \varepsilon_{j,t}, \quad (1)$$

$$\beta_{0,j} = \gamma_0 + \gamma_1 z_j + u_j, \quad (2)$$

where $x_{j,t}$ is a time-varying covariate with coefficient β_1 and $\beta_{0,j}$ is the varying intercept at the lower level, which is a function of the intercept γ_0 at the higher level and a time-invariant covariate z_j with coefficient γ_1 . $\varepsilon_{j,t}$ and u_j denote the error term at the lower level and the higher level, respectively, with the assumptions of normal distribution:

$$\varepsilon_{j,t} \sim N(0, \sigma_\varepsilon^2) \quad (3)$$

$$u_j \sim N(0, \sigma_u^2) \quad (4)$$

where σ_ε^2 and σ_u^2 denote the variance of the two distributions, respectively.

An alternative form of the RE model is to combine Equations (1) and (2) by substituting the latter into the former:

$$y_{j,t} = \gamma_0 + \beta_1 x_{j,t} + \gamma_1 z_j + u_j + \varepsilon_{j,t}. \quad (5)$$

With Equation 5, we can simultaneously estimate the effects from the micro and macro parts of the model (Bell & Jones, 2015). However, it is well-known that RE models often suffer from a violation of the exogeneity assumption, that is:

$$\text{Cov}(x_{j,t}, u_j) = 0. \quad (6)$$

To deal with this problem, we can consider that $x_{j,t}$ contains two parts: one is specific to the unit and the other represents the differences within units (Bell & Jones, 2015). That is:

$$x_{j,t} = x_j^B + x_{jt}^W. \quad (7)$$

According to Equation 7, we can see that the covariance between $x_{j,t}$ and u_j comes from the covariance between x_j^B and u_j . We can include the unit mean of $x_{j,t}$ into the higher level to account for the between effect (Mundlak, 1978). By doing so, Equation (2) is now as follows:

$$\beta_{0,j} = \gamma_0 + \gamma_1 z_j + \gamma_2 \bar{x}_j + u_j. \quad (8)$$

After combining Equation 1 and Equation 8, we have:

$$y_{j,t} = \gamma_0 + \beta_1 x_{j,t} + \gamma_1 z_j + \gamma_2 \bar{x}_j + u_j + \varepsilon_{j,t} \quad (9)$$

We can rearrange Equation 9⁶ to:

$$y_{j,t} = \gamma_0 + \beta_1 (x_{j,t} - \bar{x}_j) + \beta_2 \bar{x}_j + \gamma_1 z_j + u_j + \varepsilon_{j,t} \quad (10)$$

where β_1 is the *within* effect and β_2 is the *between* effect (Bartels, 2008; Leyland, 2010).

Through the within-between formulation of RE models, β_1 captures the temporal effect within units and β_2 captures the cross-sectional effect which is constant over time. In other words, the within-between regression model can separate the effects of the short-term and long-term factors.

Equation 11 encapsulates the hypothesized causal pattern:

$$\begin{aligned} \text{IU views}_{j,t} = & \beta_0 + \beta_1(\text{Economic evaluation}_{j,t} - \overline{\text{Economic evaluation}_j}) \\ & + \beta_2\overline{\text{Economic evaluation}_j} + \beta_3(\text{Cross-Strait}_{j,t} - \overline{\text{Cross-Strait}_j}) \\ & + \beta_4\overline{\text{Cross-Strait}_j} + \beta_5(\text{Evaluation of President}_{j,t} - \overline{\text{Evaluation of President}_j}) \\ & + \beta_6\overline{\text{Evaluation of President}_j} + \beta_7(\text{KMT feeling}_{j,t} - \overline{\text{KMT feeling}_j}) \\ & + \beta_8\overline{\text{KMT feeling}_j} + \beta_9(\text{DPP feeling}_{j,t} - \overline{\text{DPP feeling}_j}) + \beta_{10}\overline{\text{DPP feeling}_j} \\ & + \beta_{11}(\text{Pro-KMT news}_{j,t} - \overline{\text{Pro-KMT news}_j}) \\ & + \beta_{12}\overline{\text{Pro-KMT news}_j} + \beta_{13}(\text{Pro-DPP news}_{j,t} - \overline{\text{Pro-DPP news}_j}) \\ & + \beta_{14}\overline{\text{Pro-DPP news}_j} + \beta_{15}\text{sex}_j + \beta_{16}\text{age}_j + \beta_{17}\text{education}_j \\ & + \beta_{18}\text{KMT identification}_j + \beta_{19}\text{DPP identification}_j \\ & + \beta_{20}\text{Taiwanese identity}_j + \beta_{21}\text{Dual identity}_j + u_j + \epsilon_{j,t} \end{aligned} \quad (11)$$

Main results

In this section, we first show trends in IU positions through the descriptive statistics. Then, we present the results of regression analysis and examine our hypotheses.

Distribution of independence–unification views

Examining panel data from between 2001 and 2003, Tsai et al. (2008) showed that respondents changed to ‘status quo’ from ‘independence’ and ‘unification’ categories. For example, 73.8% of ‘pro-unification’ respondents switched to ‘status quo’ and 46.9% of ‘pro-independence’ respondents also did so. Before investigating the within-individual distribution, we observed the relative frequencies in 2018 and 2019, as shown in Figure 2. It is apparent that the proportion of respondents who lean toward independence increased: ‘independence later’ grew from 16.3% to 25.0% and ‘immediate independence’ grew from 5.8% to 6.6%. In contrast, the two unification-related categories shrank from 21.5% to 12.6% and two categories related to the status quo showed little change over the period. Compared to Tsai et al. (2008), the distribution seems to skew towards independence rather than a unimodal bell-shaped distribution.

Figure 3 displays the cross-tabulation of issue position in 2018 and 2019. It is apparent that most people who chose ‘status quo, decide later,’ ‘status quo forever,’ and ‘independence later’ in 2018 maintained their position in 2019. However, only 21.9% of respondents who answered ‘immediate unification’ gave the same answer in 2019. Within this category, 24.8% of respondents switched to ‘unification later,’ and 27.3% changed to

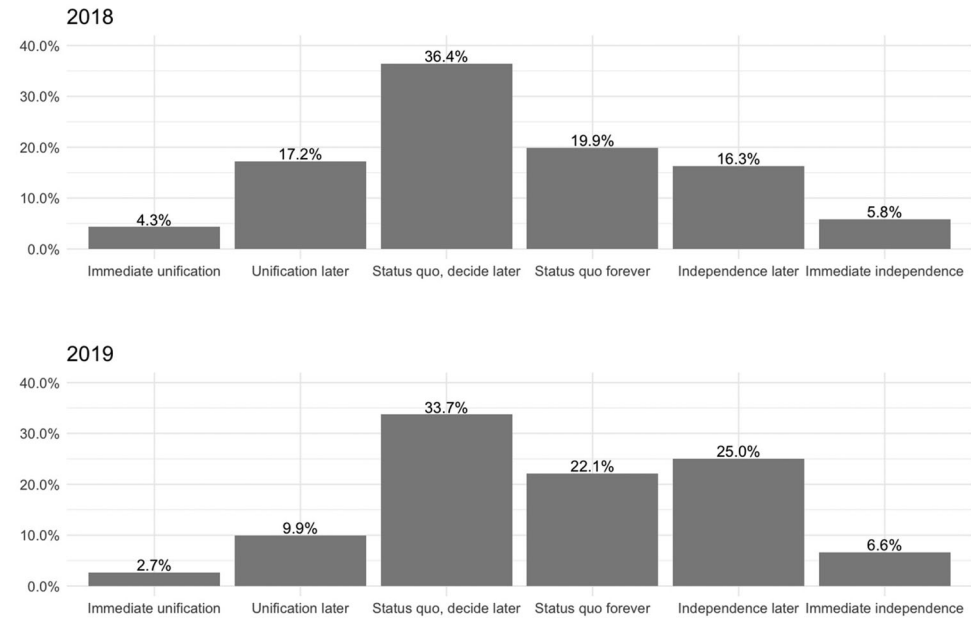


Figure 2. Independence–Unification Issue Position in 2018 and 2019. Data: Huang (2019b).

‘status quo, decide later.’ In other words, more than 50% of respondents who held ‘immediate unification’ views shifted to other views within a year. The ‘unification later’ group was more stable than ‘immediate unification’ – 32.4% of people who chose this position in 2018 maintained the same position one year later. However, 31.5% of respondents that chose ‘unification later’ switched to ‘status quo, decide later.’ Conversely, there was a higher level of stability on the pro-independence side of the IU spectrum. Within the ‘immediate independence’ category, 37.2% of respondents maintained their position and 36.3% switched to ‘independence later.’ The ‘independence later’ group was the most stable – 62.5% of respondents that answered ‘independence later’ in 2018 did not change their views in 2019.

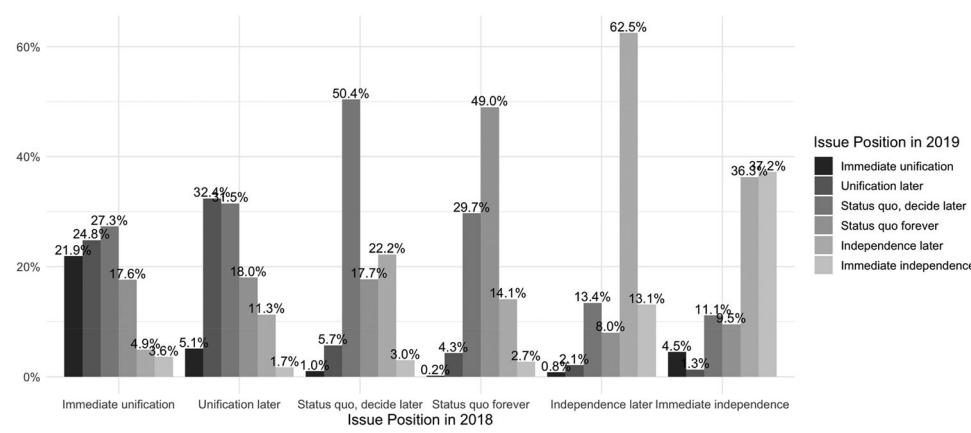


Figure 3. Change in Independence–Unification Issue Position, 2018 and 2019. Data: Huang (2019b).

Overall, Figures 2 and 3 show that the centrist position lost ground between 2018 and 2019, while support for independence increased substantially over the same period. The evident change in this important issue merits in-depth analysis.

The analysis of within-between model

To explain the trends in IU views, we differentiated short-term effects from long-term effects by using the within-between regression model. The results of analysis are summarized as follows. We first looked at the *within* effects of several explanatory variables on IU views. First, positive evaluation of the national economy and President Tsai's performance in cross-Straits relations was positively correlated with the independence position. Watching pro-DPP television news was also associated with pro-independence views. However, the correlation between presidential approval and the IU positional stance was not significant. Moreover, we did not observe a significant relationship between partisanship and the IU positional stance. These results suggest that, when respondents think that the national economy is getting better, they tend to favor independence. By the same token, when respondents think that President Tsai's performance on cross-Straits relations is getting better, they also tend to favor independence.

The second panel of Table 1 shows the *between* effects, that is, the effects of individuals, on IU views. First, individuals with positive evaluations of the national economy favor independence. Second, individuals who are satisfied with President Tsai's dealing with cross-Straits relations favor independence. Third, voters who approve of President Tsai are less supportive of independence. Finally, voters who support KMT favor unification while those who support DPP favor independence. Taiwanese identity and dual identity also affect IU views.

Table 1. Within-between regression model of IU views.

Within Effects	Est.	S.E.	t val.	d.f.	p
Economic evaluations	0.11	0.04	2.87	1370.24	0.00
Cross-Straits relations	0.24	0.05	4.96	1368.40	0.00
Tsai's approval	0.05	0.05	0.87	1367.67	0.38
KMT rating	-0.01	0.02	-0.41	1394.28	0.68
DPP rating	0.01	0.02	0.39	1412.37	0.70
Pro-KMT TV news	-0.00	0.03	-0.03	1365.97	0.97
Pro-DPP TV news	0.07	0.03	2.53	1363.68	0.01
Between Effects	Est.	S.E.	t val.	d.f.	p
Economic evaluations	0.14	0.04	3.38	1374.64	0.00
Cross-Straits relations	0.26	0.06	4.48	1391.94	0.00
Tsai's approval	-0.17	0.06	-3.05	1362.69	0.00
KMT rating	-0.12	0.01	-7.98	1534.00	0.00
DPP rating	0.06	0.02	3.64	1466.08	0.00
Pro-KMT TV news	-0.04	0.04	-1.02	1374.17	0.31
Pro-DPP TV news	0.01	0.03	0.34	1365.30	0.73
(Intercept)	2.78	0.22	12.85	1684.05	0.00
Female	0.07	0.05	1.51	1371.29	0.13
Age	-0.00	0.00	-0.35	1475.05	0.72
Education	0.02	0.02	0.69	1513.27	0.49
Pan-Blue identification	-0.06	0.06	-1.01	2724.92	0.31
Pan-Green identification	0.10	0.06	1.68	2727.86	0.09
Taiwanese identity	0.68	0.11	6.44	2737.66	0.00
Dual identity	0.21	0.10	2.07	2717.32	0.04

Note: $N = 1380$. Model fit: $AIC = 8179.15$, $BIC = 8321.31$, $Pseudo-R^2$ (fixed effects) = 0.27, $Pseudo-R^2$ (total) = 0.52, Entity ICC = 0.35.

Data: Huang (2019b).

Our five hypotheses are mostly supported by the evidence presented in Table 1. Voters who feel positively about the national economy believe that economic cooperation between Taiwan and China is not necessary. Therefore, they do not favor unification. By the same token, voters who think that President Tsai has handled cross-Straits relations well would think a further step toward unification is not necessary. However, the result that positive presidential approval is associated with leaning toward unification is surprising. We do not have a good explanation so far. Finally, the relationship between partisanship and IU issue position is as expected.

The dependent variable, a 6-item measurement of IU position, is widely used as either a discrete or continuous variable to measure IU views. However, in order to ensure that our findings are sustained when a different dependent variable is used, we also reran our within-between model using a 0–10-point scale where 0 means independence and 10 means unification. The results showed larger coefficients but in the same direction as the 6-item measurement. President Tsai's overall approval did not have within or between effects on the new outcome variable, but a positive impression of the KMT and DPP both become significant predictors. The estimates are shown in Appendix B.

In summary, the results show that the *between* effects of the five variables mentioned above are correlated with the IU positional stance are consistent with findings in literature. Regarding the *within* effects, voters may change their evaluation about the national economy and President Tsai's performance regarding cross-Straits relations in the short term. These changes further influence their attitudes towards the IU issue. Specifically, when they think that the national economy has improved and have become satisfied with President Tsai's performance regarding cross-Straits relations, they lean more towards independence. In contrast, voters' approval of President Tsai and partisanship do not have a short-term effect on the IU positional stance.

Conclusion

This study applied a within-between model to examine the extent to which IU views are influenced by short-term evaluations of the national economy and the president's handling of cross-Straits relations. We assume that a certain proportion of people will change their IU views from unification to the status quo or independence in response to international or domestic events, while only a few people will move in the opposite direction from independence to unification. Our preliminary results show that Taiwanese people indeed change their IU views due to both the external and domestic environment. People who perceive that the economy has improved and whose views of President Tsai have become more favorable tend to flock towards the independence side. People who approve President Tsai's handling of cross-Straits relations also move toward independence. If we only consider difference between individuals, feelings about the KMT and DPP also stand out as significant predictors of the issue position.

The primary implication of this study is that IU views are fluid – people may decide their political attitudes based on the performance of the president and evaluations of the national economy. Our research offers a different explanation for the formation of Taiwanese IU views. According to Huang (2019), younger cohorts and the Taiwan-centered high school curriculum implemented in 1999 have led to a shift in IU views. In other words, the evolution of IU views has its roots in socialization rather than events such

as China's missile tests in the Taiwan Strait between July 1995 and March 1996. Our findings instead echo the 'rational public' theory proposed by Page and Shapiro (1992), in which the mass public respond to international events and domestic issues in a predictable way (Shapiro & Page, 1988). We found that the president's response to major events triggered shifts in IU views. Another contribution of this research lies in the effect of watching partisan television news. Changes in individuals' media exposure is related to IU views. This finding implies that public opinion responds to the dynamics of media consumption. In this case, it is necessary to include the media variables in the model of public opinion formation.

In our within-between model, we only measured self-reported media exposure using television news because the importance of newspapers is declining. We cannot rule out other media that may come into play. For example, many people read, discuss, or share news on social media. The content of social media is much more diverse and diffuse than television news programs. Despite this, news on social media deserves more attention as an explanatory variable for political attitudes.

This study argued that people's satisfaction with the president's handling of cross-Strait relations influences their IU views. However, it is also theoretically possible that the causality runs in the opposite direction. Like all of observational data, this causal direction problem is difficult to solve empirically. By the same token, media exposure may also be endogenous to IU views. If there is simultaneous causality, the effects of the above two factors on IU views might be overestimated in our analysis. In the future, we may adopt experimental designs to correct for the endogeneity problem to advance our understanding of public opinion. For example, during experiments, we could allow subjects to decide how much partisan media they want to watch (Della Vigna & Kaplan, 2007). Having said that, this research has integrated the concept of the rally 'round the flag effect and media exposure perspective to shed some light on attitude change.

Notes

1. The Media Agency Association is made up of Taiwan-based advertising companies. It publishes a report on the advertising and media market every year. Please see the website for more details. <https://maataipei.org>.
2. We compiled data points from a series of telephone and mobile phone interviews on presidential satisfaction, part of Taiwan's Election and Democratization Study (TEDS), 2016–2020 (III). The coordinator of multi-year project TEDS is Professor Chi Huang (National Chengchi University). More information is available on the TEDS website (<http://teds.nccu.edu.tw/>). The authors appreciate the assistance in providing data by the institute and individuals aforementioned. The authors are alone responsible for views expressed herein.
3. The DPP had won every Kaohsiung mayoral election since 1998. In 2014, the DPP mayor, Chen Chu, was re-elected with 68 percent of the vote.
4. The KMT's post-election review indicated that the "sense of the nation's impending doom" was the top reason why the KMT lost to the DPP. See <http://www.taipeitimes.com/News/front/archives/2020/01/16/2003729327>.
5. <https://www.taiwannews.com.tw/en/news/3862410>.
6. γ_2 in Equation 9 models the difference between the within and between effects, that is, $\gamma_2 = \beta_2 - \beta_2$ (Berlin, Kimmel, Ten Have, & Sammel, 1999).

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No potential conflict of interest was reported by the author(s).

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Appendix A

Table A1. Variables and operationalization.

Name	Variable	Question	Options
TONDU	Unification-independence position	Concerning the relationship between Taiwan and mainland China, which of the following six positions do you agree with: (1) immediate unification, (2) immediate independence, (3) maintain the status quo, move toward unification in the future, (4) maintain the status quo, move toward independence in the future, (5) maintain the status quo, decide either unification or independence in the future, (6) maintain the status quo forever	'Immediate unification' = 1, 'Unification later' = 2, 'Status quo, decide later' = 3, 'Status quo forever' = 4, 'Independence later' = 5, 'Immediate independence' = 6
vPF1 (2018), vPE1(2019)	Unification-independence scale	In our society people often talk about the question of Taiwan independence from or unification with China. Some people say that Taiwan should declare independence right away. Other people say that Taiwan and China should unify right away. Yet other people have opinions between these two positions. On this card, the position that Taiwan should immediately declare independence is at 0 on a scale from 0 to 10, and the position that Taiwan should immediately unify with the mainland is at 10. About where on this scale does your own view lie?	'Immediate independence' = 0, 'Immediate unification' = 10
vPD1(2018); vPD1(2019)	Tsai's approval	How satisfied are you with Tsai Ing-wen's overall performance as president since her inauguration?	'Not satisfied at all' = 1, 'Not very satisfied' = 2, 'Satisfied' = 3, 'Very satisfied' = 4
vPD9(2018); vPD9(2019);	Evaluations of cross-Straits relations	How satisfied are you with Tsai Ing-wen's performance in handling the Cross-strait relations?	'Not satisfied at all' = 1, 'Not very satisfied' = 2, 'Satisfied' = 3, 'Very satisfied' = 4
vPC1(2018); vPC1(2019);	Evaluations of national economy	Would you say that over the past six months, the state of economy of Taiwan has gotten better, stayed about the same, or gotten worse?	'Get much worse' = 1, 'Get worse' = 2 'Stay the same' = 3, 'Get better' = 4, 'Get much better' = 5
vGA1(2018); vGA1(2019)	KMT rating	The point that you have no positive impression about a party is 0 on a scale from 0 to 10, and the point that you have very positive impression is 10. About where on this scale does your impression about KMT lie?	'No positive impression' = 0, 'Very positive impression' = 10
vGA2(2018); vGA2(2019)	DPP rating	The point that you have no positive impression about a party is 0 on a scale from 0 to 10, and the point that you have very positive impression is 10.	'No positive impression' = 0, 'Very positive impression' = 10

(Continued)

Table A1. Continued.

Name	Variable	Question	Options
vNE1b1m01- vNE1b1m80 (2018), vNE1b1m01- vNE1b1m80; vNE2_2m1-vNE2_2m64 (2019)	Pro-DPP TV	About where on this scale does your impression about DPP lie? Which TV channel do you obtain news about politics or public affairs from?	Formosa TV, Sanlih Entertainment Television, Era TV, NEXT TV, Chinese Television Station (CTS) = 1, otherwise = 0
vNE1b1m01- vNE1b1m80 (2018), vNE1b1m01- vNE1b1m80; vNE2_2m1-vNE2_2m64 (2019)	Pro-KMT TV	Which TV channel do you obtain news about politics or public affairs from?	TVBS, CTI Television, China Television Company = 1, otherwise = 0
vPJ1(2018); vPI1(2019)	Taiwanese identity	Do you think of yourself as Taiwan, Chinese, or both?	'Taiwanese' = 1, otherwise = 0
vPJ1(2018); vPI1(2019)	Dual identity	Do you think of yourself as Taiwan, Chinese, or both?	'Both Taiwanese and Chinese' = 1, otherwise = 0
PARTYID	Party Identification	Political parties in Taiwan have their own supporters. Among these political parties, which one do you support?	Dummy variables: Pan-blue (KMT, People First Party); Pan-green (DPP, New Power Party)
vP7(2018);vP6(2019)	Education	What is your education level?	illiterate, primary school, junior high school, high school, and some college
vP2gp2a(2018)	Age	When were you born ?	18–95 years old
SEX	Gender		Male = 0, Female = 1

Appendix B

Table B1. Within-Between Regression Model of IU Views (D.V. = 0–10 position on independence – unification scale).

Within Effects	Est.	S.E.	<i>t</i> val.	d.f.	<i>p</i>
Economic evaluations	−0.31	0.07	−4.61	1339.57	0.00
Cross-Strait relations	−0.28	0.08	−3.44	1339.54	0.00
Tsai's approval	−0.05	0.09	−0.60	1338.64	0.55
KMT rating	0.08	0.03	2.97	1363.36	0.00
DPP rating	−0.05	0.03	−1.90	1382.35	0.06
Pro-KMT TV news	0.09	0.06	1.47	1336.66	0.14
Pro-DPP TV news	−0.10	0.05	−2.07	1334.03	0.04
Between Effects					
Economic evaluations	−0.16	0.08	−2.14	1345.40	0.03
Cross-Strait relations	−0.37	0.10	−3.55	1361.65	0.00
Tsai's approval	0.10	0.10	1.00	1332.87	0.32
KMT rating	0.27	0.03	10.34	1505.22	0.00
DPP rating	−0.16	0.03	−5.24	1440.84	0.00
Pro-KMT TV news	−0.00	0.07	−0.01	1344.48	0.99
Pro-DPP TV news	−0.11	0.06	−1.90	1337.05	0.06
(Intercept)	5.93	0.39	15.23	1657.63	0.00
Female	0.03	0.09	0.31	1341.81	0.75
Age	0.00	0.00	1.04	1451.40	0.30
Education	0.01	0.04	0.38	1499.37	0.71
Pan-Blue identification	−0.06	0.10	−0.58	2699.60	0.56
Pan-Green identification	−0.08	0.11	−0.76	2700.00	0.45
Taiwanese identity	−1.31	0.19	−7.07	2685.11	0.00
Dual identity	−0.61	0.18	−3.42	2637.83	0.00

Note: $N = 1361$. Model fit: AIC = 11,064.08, BIC = 11,205.9, Pseudo- R^2 (fixed effects) = 0.30, Pseudo- R^2 (total) = 0.58, Entity ICC = 0.43.

Data: Huang (2019b).